Chapter 15

# The Lie Correspondence and the Exponential Map

**Road map**

The bridge from a matrix Lie algebra to its matrix Lie group is the exponential map. We first prove that the matrix-exponential series converges and establish its basic identities, including $\det(\mathrm{e}^X) = \mathrm{e}^{\operatorname{tr} X}$. We then construct the local matrix logarithm, classify all one-parameter subgroups of $\mathrm{GL}(n, \mathbb{K})$, and prove the Lie–Trotter product formula. After an application to quantum statistical mechanics, we define the Lie algebra of a closed matrix group and prove its vector-space, bracket, conjugation, and tangent-space properties. The final sections explain local and global surjectivity of the exponential map, products of exponentials in connected groups, and the relation between a group and its identity component. Here and throughout, $\mathbb{K}$ means either $\mathbb{R}$ or $\mathbb{C}$.

## 15.1 The matrix exponential

### 15.1.1 Definition and norm convergence

**Definition 15.1: Matrix exponential**

For $X \in \mathrm{Mat}_n(\mathbb{K})$, define

$$\boxed{\exp X = \mathrm{e}^X := \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots .} \tag{15.1}$$

Thus the matrix exponential is initially a map

$$\exp : \mathrm{Mat}_n(\mathbb{K}) \longrightarrow \mathrm{Mat}_n(\mathbb{K}).$$

We shall prove that its values are automatically invertible, so its actual codomain may be taken to be $\mathrm{GL}(n, \mathbb{K})$.

To make the infinite series rigorous, put on $\mathrm{Mat}_n(\mathbb{K})$ the Frobenius, or Hilbert–Schmidt, norm

$$\|X\|_F := \sqrt{\operatorname{tr}(X^\dagger X)} = \left(\sum_{i,j=1}^{n} |X_{ij}|^2\right)^{1/2}.$$

For a real matrix $X^\dagger = X^T$. This is indeed a norm:

$$\|X+Y\|_F \le \|X\|_F + \|Y\|_F, \qquad \|\lambda X\|_F = |\lambda|\,\|X\|_F,$$

and $\|X\|_F \ge 0$, with equality only for $X = 0$.

The property needed for multiplying power series is submultiplicativity.

**Lemma 15.1: Submultiplicativity of the Frobenius norm**

For $X, Y \in \mathrm{Mat}_n(\mathbb{K})$,

$$\|XY\|_F \le \|X\|_F\,\|Y\|_F. \tag{15.2}$$

证明. Let $x_i$ be the $i$-th row of $X$, and let $y_j$ be the $j$-th column of $Y$. With the standard Hermitian inner product on $\mathbb{K}^n$,

$$(XY)_{ij} = \sum_{k=1}^{n} X_{ik} Y_{kj} = \langle x_i^*, y_j \rangle.$$

The Cauchy–Schwarz inequality gives

$$|(XY)_{ij}|^2 \le \|x_i\|_{\mathbb{K}^n}^2 \|y_j\|_{\mathbb{K}^n}^2.$$

Summing first over $i, j$ and then separating the two finite sums,

$$\begin{aligned}
\|XY\|_F^2 &= \sum_{i,j} |(XY)_{ij}|^2 \\
&\le \sum_{i,j} \|x_i\|_{\mathbb{K}^n}^2 \|y_j\|_{\mathbb{K}^n}^2 \\
&= \left(\sum_i \|x_i\|_{\mathbb{K}^n}^2\right)\left(\sum_j \|y_j\|_{\mathbb{K}^n}^2\right) \\
&= \|X\|_F^2 \|Y\|_F^2.
\end{aligned}$$

Taking square roots proves (15.2). ■

**Theorem 15.1: The exponential series is well defined**

For every $X \in \mathrm{Mat}_n(\mathbb{K})$, the series (15.1) converges absolutely in the Frobenius norm. Moreover,

$$\boxed{\|\mathrm{e}^X\|_F \le \mathrm{e}^{\|X\|_F}.} \tag{15.3}$$

证明. Repeated use of (15.2) gives $\|X^k\|_F \le \|X\|_F^k$. Hence

$$\sum_{k=0}^{\infty} \left\| \frac{X^k}{k!} \right\|_F \le \sum_{k=0}^{\infty} \frac{\|X\|_F^k}{k!} = \mathrm{e}^{\|X\|_F} < \infty.$$

The vector space of $n \times n$ matrices is finite dimensional and therefore complete. Absolute convergence consequently implies convergence. The same estimate, together with the triangle inequality, proves (15.3). ■

### 15.1.2 First algebraic properties

**Proposition 15.1: Commuting exponential law**

If $X, Y \in \mathrm{Mat}_n(\mathbb{K})$ commute, then

$$\mathrm{e}^{X+Y} = \mathrm{e}^X \mathrm{e}^Y = \mathrm{e}^Y \mathrm{e}^X. \tag{15.4}$$

In particular,

$$(\mathrm{e}^X)^{-1} = \mathrm{e}^{-X},$$

so $\mathrm{e}^X \in \mathrm{GL}(n, \mathbb{K})$ for every $X$.

证明. When $XY = YX$, the ordinary binomial formula remains valid:

$$(X+Y)^k = \sum_{r=0}^{k} \binom{k}{r} X^r Y^{k-r}.$$

Absolute convergence allows the terms to be regrouped. Therefore

$$\begin{aligned} \mathrm{e}^{X+Y} &= \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{r=0}^{k} \binom{k}{r} X^r Y^{k-r} \\ &= \sum_{r,s \geq 0} \frac{X^r Y^s}{r!\, s!} = \left( \sum_{r=0}^{\infty} \frac{X^r}{r!} \right) \left( \sum_{s=0}^{\infty} \frac{Y^s}{s!} \right). \end{aligned}$$

This is $\mathrm{e}^X \mathrm{e}^Y$. Taking $Y = -X$, which certainly commutes with $X$, gives

$$\mathrm{e}^{-X} \mathrm{e}^X = \mathrm{e}^0 = I = \mathrm{e}^X \mathrm{e}^{-X}.$$

■

**Why commutativity matters**

For arbitrary matrices it is generally false that $\mathrm{e}^{X+Y} = \mathrm{e}^X \mathrm{e}^Y$. The missing information is measured, to first nontrivial order, by the commutator $[X, Y] = XY - YX$. The Lie–Trotter formula in Section 15.4 will recover $\mathrm{e}^{X+Y}$ without assuming that $X$ and $Y$ commute.

**Proposition 15.2: Equivariance under similarity**

For $A \in \mathrm{GL}(n, \mathbb{K})$ and $X \in \mathrm{Mat}_n(\mathbb{K})$,

$$\boxed{\mathrm{e}^{AXA^{-1}} = A \mathrm{e}^X A^{-1}.} \tag{15.5}$$

证明. Induction gives $(AXA^{-1})^k = AX^kA^{-1}$. Substituting this into the series,

$$e^{AXA^{-1}} = \sum_{k=0}^{\infty} \frac{AX^kA^{-1}}{k!} = A\left(\sum_{k=0}^{\infty} \frac{X^k}{k!}\right) A^{-1}.$$

■

**Theorem 15.2: Determinant of an exponential**

For every $X \in \mathrm{Mat}_n(\mathbb{K})$,

$$\boxed{\det(e^X) = e^{\operatorname{tr} X}.} \tag{15.6}$$

证明. First suppose that $X$ is diagonalizable over $\mathbb{C}$:

$$X = SDS^{-1}, \qquad D = \operatorname{diag}(d_1, \dots, d_n).$$

Then (15.5) yields

$$\det(e^X) = \det(Se^DS^{-1}) = \det(e^D) = \prod_{j=1}^{n} e^{d_j} = e^{\sum_j d_j} = e^{\operatorname{tr} X}.$$

The handwritten argument then observes that complex-diagonalizable matrices are dense and that both sides are continuous, which extends the equality to all matrices.

For completeness, one can avoid the density step by using complex Schur triangularization. Every complex matrix has the form $X = UTU^\dagger$, where $U$ is unitary and $T$ is upper triangular. The matrix $e^T$ is upper triangular with diagonal entries $e^{T_{11}}, \dots, e^{T_{nn}}$; hence

$$\det(e^X) = \det(e^T) = \prod_j e^{T_{jj}} = e^{\operatorname{tr} T} = e^{\operatorname{tr} X}.$$

A real matrix may be regarded as a complex matrix, so the same calculation also proves the real case. ■

## 15.2 The local matrix logarithm

**Definition 15.2: Logarithm near the identity**

If $A \in \mathrm{Mat}_n(\mathbb{K})$ satisfies $\|A - I\|_F < 1$, define

$$\boxed{\log A := \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}(A - I)^k.} \tag{15.7}$$

This is the matrix version of

$$\log(1 + a) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} a^k, \qquad |a| < 1.$$

**Proposition 15.3: Convergence and inverse identity**

If $\|A - I\|_F < 1$, the series (15.7) converges absolutely and

$$e^{\log A} = A. \tag{15.8}$$

证明. Put $B = A - I$. Then

$$\sum_{k=1}^{\infty} \left\| \frac{(-1)^{k+1}}{k} B^k \right\|_F \le \sum_{k=1}^{\infty} \frac{\|B\|_F^k}{k} < \infty.$$

Thus the logarithm converges in norm.

All terms involved are powers of the single matrix $B$, so they belong to the commutative subalgebra generated by $B$. Within that commutative algebra, absolutely convergent power series may be composed exactly as scalar power series. The scalar analytic identity $\exp(\log(1 + z)) = 1 + z$ for $|z| < 1$ therefore gives

$$\exp(\log(I + B)) = I + B = A.$$

This argument also explains why diagonalizing $A$ is unnecessary. In particular, it remains valid for a non-diagonalizable matrix. ∎

**Correction to a tempting diagonalization argument**

One may verify (15.8) first for diagonalizable matrices and then use density and continuity. However, for a non-unitary similarity $A = SDS^{-1}$, the inequality $\|D - I\|_F \le \|A - I\|_F$ need not hold. The power-series proof above is both shorter and fully rigorous. Over $\mathbb{C}$, the logarithm used here is the local branch determined by the series; for scalar eigenvalues it agrees with the principal logarithm whenever that branch is defined.

The exponential also has a local inverse in the other direction. Indeed,

$$\|e^X - I\|_F \le \sum_{k=1}^{\infty} \frac{\|X\|_F^k}{k!} = e^{\|X\|_F} - 1.$$

Consequently, if $\|X\|_F < \log 2$, then $\|e^X - I\|_F < 1$, so $\log(e^X)$ is defined. By composing the scalar series inside the commutative algebra generated by $X$,

$$\log(e^X) = X \qquad \text{for all sufficiently small } X. \tag{15.9}$$

**Theorem 15.3: Local exponential chart**

There are neighbourhoods $\mathcal{U}$ of $0$ in $\mathrm{Mat}_n(\mathbb{K})$ and $\mathcal{V}$ of $I$ in $\mathrm{GL}(n, \mathbb{K})$ for which

$$\exp : \mathcal{U} \longrightarrow \mathcal{V}$$

is a bijection with inverse log. One may take $\mathcal{U}$ inside $\{X : \|X\|_F < \log 2\}$ and $\mathcal{V}$ inside $\{A : \|A - I\|_F < 1\}$.

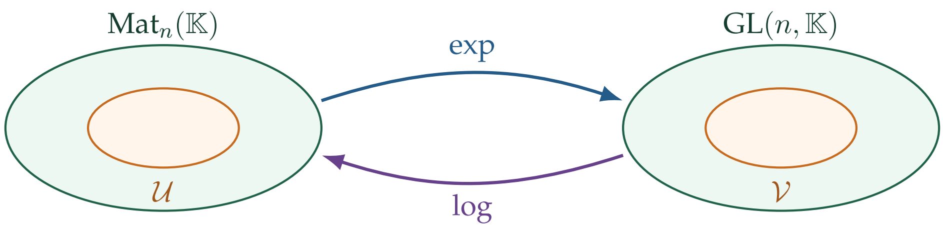


图 **15.1:** The exponential and logarithm are inverse maps near $0$ and $I$. They are not generally inverse on the whole spaces.

## 15.3 One-parameter subgroups

**Definition 15.3: One-parameter subgroup**

Let $G$ be a topological group. A *one-parameter subgroup* of $G$ is a continuous group homomorphism

$$c : (\mathbb{R}, +) \longrightarrow G.$$

Equivalently,

$$c(t + t') = c(t)c(t'), \qquad c(0) = e_G. \tag{15.10}$$

Its image $\{c(t) : t \in \mathbb{R}\}$ is an Abelian subgroup of $G$, because

$$c(t)c(t') = c(t + t') = c(t' + t) = c(t')c(t).$$

If $c(t)$ is close enough to the identity matrix, define

$$X(t) := \log c(t).$$

Since $c(t)$ and $c(t')$ commute, the local logarithm converts their product into a sum whenever all three matrices remain in the logarithm neighbourhood:

$$X(t + t') = \log(c(t)c(t')) = \log c(t) + \log c(t') = X(t) + X(t'). \tag{15.11}$$

Thus the local logarithm turns the multiplicative homomorphism law into an additive one.

**Example 15.1: The subgroup generated by a matrix**

For any $X \in \mathrm{Mat}_n(\mathbb{K})$, set

$$c_X(t) := \mathrm{e}^{tX}. \tag{15.12}$$

Because $tX$ and $t'X$ commute,

$$c_X(t)c_X(t') = \mathrm{e}^{tX}\mathrm{e}^{t'X} = \mathrm{e}^{(t+t')X} = c_X(t + t').$$

Moreover,

$$c_X(0) = I, \qquad \dot{c}_X(t) = X\mathrm{e}^{tX} = \mathrm{e}^{tX}X, \qquad \dot{c}_X(0) = X.$$

Hence $c_X$ is not only continuous but infinitely differentiable.

**Theorem 15.4: Classification of one-parameter matrix subgroups**

Every continuous one-parameter subgroup

$$c : \mathbb{R} \longrightarrow \mathrm{GL}(n, \mathbb{K})$$

has the form

$$\boxed{c(t) = \mathrm{e}^{tX} \quad (t \in \mathbb{R})} \tag{15.13}$$

for a unique $X \in \mathrm{Mat}_n(\mathbb{K})$. In particular, every continuous one-parameter subgroup is automatically smooth.

证明. **Uniqueness.** If $c(t) = \mathrm{e}^{tX}$, differentiation at $t = 0$ gives

$$X = \dot{c}(0).$$

Thus at most one matrix $X$ can work.

**Existence near** $t = 0$**.** Continuity and $c(0) = I$ allow us to choose $t_0 > 0$ so small that $\log c(t)$ is defined for $|t| \le t_0$. Put

$$X := \frac{1}{t_0} \log c(t_0).$$

Then $c(t_0) = \mathrm{e}^{t_0 X}$.

We next determine the values at dyadic subdivisions of $t_0$. The homomorphism law gives

$$c(t_0/2)^2 = c(t_0) = \mathrm{e}^{t_0 X}.$$

Both $c(t_0/2)$ and $\mathrm{e}^{t_0 X/2}$ lie in the neighbourhood on which log is the inverse of exp. Applying the local logarithm, or equivalently using uniqueness of the sufficiently small square root, gives

$$c(t_0/2) = \mathrm{e}^{t_0 X/2}.$$

Repeating the argument,

$$c(t_0/2^k) = \mathrm{e}^{t_0 X/2^k} \qquad (k \in \mathbb{N}).$$

For every integer $m$ with $0 \le m \le 2^k$,

$$c(mt_0/2^k) = c(t_0/2^k)^m = \mathrm{e}^{mt_0 X/2^k}.$$

The dyadic points $\{mt_0/2^k\}$ are dense in $[0, t_0]$. Both $c$ and $t \mapsto \mathrm{e}^{tX}$ are continuous, hence

$$c(t) = \mathrm{e}^{tX} \qquad (0 \le t \le t_0).$$

The equality for $-t_0 \le t \le 0$ follows from

$$c(-t) = c(t)^{-1} = \mathrm{e}^{-tX}.$$

**Extension to every real** $t$**.** Given $t \in \mathbb{R}$, choose $N \in \mathbb{N}$ with $|t/N| \le t_0$. Then

$$c(t) = c(t/N)^N = (\mathrm{e}^{(t/N)X})^N = \mathrm{e}^{tX}.$$

This proves existence and completes the proof. ∎

**A useful conceptual consequence**

For matrix groups, continuity of a homomorphism $\mathbb{R} \to \mathrm{GL}(n, \mathbb{K})$ is much stronger than it first appears: it forces analyticity. No separate differentiability assumption is needed.

## 15.4 The Lie–Trotter product formula

**Theorem 15.5: Lie–Trotter product formula**

For arbitrary $X, Y \in \mathrm{Mat}_n(\mathbb{K})$, with no commutativity assumption,

$$\boxed{\mathrm{e}^{X+Y} = \lim_{m\to\infty} \left(\mathrm{e}^{X/m}\mathrm{e}^{Y/m}\right)^m.} \tag{15.14}$$

The limit is in any matrix norm, and therefore entry by entry as well.

证明. For large $m$, expand each exponential to second order in $m^{-1}$:

$$\mathrm{e}^{X/m} = I + \frac{X}{m} + O(m^{-2}), \qquad \mathrm{e}^{Y/m} = I + \frac{Y}{m} + O(m^{-2}).$$

The notation $O(m^{-2})$ means a matrix whose norm is at most $C/m^2$ for a constant $C$ independent of sufficiently large $m$. Multiplying,

$$\mathrm{e}^{X/m}\mathrm{e}^{Y/m} = I + \frac{X+Y}{m} + O(m^{-2}). \tag{15.15}$$

This matrix approaches $I$, so the local logarithm is defined for all sufficiently large $m$. Since $\log(I + E) = E + O(\|E\|^2)$,

$$\log\left(\mathrm{e}^{X/m}\mathrm{e}^{Y/m}\right) = \frac{X+Y}{m} + O(m^{-2}).$$

Exponentiating after multiplying by $m$,

$$\begin{aligned}\left(\mathrm{e}^{X/m}\mathrm{e}^{Y/m}\right)^m &= \exp\left[m\log\left(\mathrm{e}^{X/m}\mathrm{e}^{Y/m}\right)\right] \\ &= \exp\left(X + Y + O(m^{-1})\right).\end{aligned}$$

The matrix exponential is continuous, so the right-hand side tends to $\mathrm{e}^{X+Y}$. ∎

**Why the formula is important**

Equation (15.14) separates a sum of noncommuting generators into many alternating short evolutions. This is a basic tool in quantum Monte Carlo methods, imaginary-time path integrals, operator splitting, and the numerical solution of differential equations.

### 15.4.1 Application: free energy and imaginary-time response

The handwritten lecture applies the product formula to a standard problem from quantum statistical mechanics. Let

$$H(\lambda) = H_0 + \lambda A, \qquad \beta := \frac{1}{k_{\mathrm{B}}T},$$

where $H_0$ and $A$ are Hermitian matrices. They need not commute. The partition function and free energy are

$$Z(\lambda) = \operatorname{tr} \mathrm{e}^{-\beta H(\lambda)}, \qquad F(\lambda) = -k_{\mathrm{B}}T \log Z(\lambda) = -\frac{1}{\beta} \log Z(\lambda). \tag{15.16}$$

For any observable $B$, its thermal expectation value at the chosen $\lambda$ is

$$\langle B \rangle_\lambda := \frac{1}{Z(\lambda)} \operatorname{tr}\Big(B\mathrm{e}^{-\beta H(\lambda)}\Big).$$

The derivative of a noncommutative exponential cannot be found by simply moving $H'(\lambda)$ through the exponential. The correct finite-dimensional Duhamel formula is

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathrm{e}^{M(\lambda)} = \int_0^1 \mathrm{e}^{(1-s)M(\lambda)} M'(\lambda)\mathrm{e}^{sM(\lambda)}\,\mathrm{d}s. \tag{15.17}$$

It follows either by differentiating the power series and regrouping the possible insertion positions of $M'$, or by applying the Lie–Trotter formula to $\mathrm{e}^{M(\lambda)+\delta M'(\lambda)}$ and taking $\delta \to 0$.

For $M = -\beta H$, formula (15.17) becomes

$$\frac{\partial}{\partial\lambda}\mathrm{e}^{-\beta H} = -\int_0^\beta \mathrm{e}^{-(\beta-\tau)H} A \mathrm{e}^{-\tau H}\,\mathrm{d}\tau. \tag{15.18}$$

Taking the trace and using cyclic invariance,

$$\operatorname{tr}(BCD) = \operatorname{tr}(CDB) = \operatorname{tr}(DBC),$$

gives

$$\begin{aligned} Z'(\lambda) &= -\int_0^\beta \operatorname{tr}\Big(\mathrm{e}^{-(\beta-\tau)H} A \mathrm{e}^{-\tau H}\Big)\,\mathrm{d}\tau \\ &= -\int_0^\beta \operatorname{tr}\Big(A\mathrm{e}^{-\beta H}\Big)\,\mathrm{d}\tau \\ &= -\beta \operatorname{tr}\Big(A\mathrm{e}^{-\beta H}\Big). \end{aligned}$$

Consequently,

$$\boxed{\frac{\partial F}{\partial\lambda} = -\frac{1}{\beta}\frac{Z'}{Z} = \langle A \rangle_\lambda.} \tag{15.19}$$

This conclusion is valid even when $[H_0, A] \neq 0$. The cyclic invariance of the trace is precisely what makes the first derivative look classical.

For the second derivative, define the imaginary-time Heisenberg operator

$$A(\tau) := \mathrm{e}^{\tau H} A \mathrm{e}^{-\tau H}, \qquad 0 \le \tau \le \beta. \tag{15.20}$$

Differentiating $\langle A\rangle = Z^{-1}\,\mathrm{tr}(A\mathrm{e}^{-\beta H})$, and using (15.18), yields

$$\begin{aligned}\frac{\partial^2 F}{\partial\lambda^2} &= -\frac{Z'}{Z^2}\,\mathrm{tr}(A\mathrm{e}^{-\beta H}) + \frac{1}{Z}\,\mathrm{tr}\Big(A\frac{\partial}{\partial\lambda}\mathrm{e}^{-\beta H}\Big)\\ &= \beta\langle A\rangle^2 - \frac{1}{Z}\int_0^\beta \mathrm{tr}\Big(A\mathrm{e}^{-(\beta-\tau)H}A\mathrm{e}^{-\tau H}\Big)\,\mathrm{d}\tau\\ &= \boxed{\beta\langle A\rangle^2 - \int_0^\beta \langle A(\tau)A(0)\rangle\,\mathrm{d}\tau.}\end{aligned} \tag{15.21}$$

The integral is the imaginary-time, or Kubo–Mori, correlation. This is the thermodynamic response quantity requested in the handwritten calculation.

If $A$ commutes with $H$, then $A(\tau) = A$, and (15.21) reduces to

$$\frac{\partial^2 F}{\partial\lambda^2} = \beta\langle A\rangle^2 - \beta\langle A^2\rangle = -\beta\big(\langle A^2\rangle - \langle A\rangle^2\big). \tag{15.22}$$

Thus, for the convention $H = H_0 + \lambda A$, the curvature is minus $\beta$ times the variance in the commuting case.

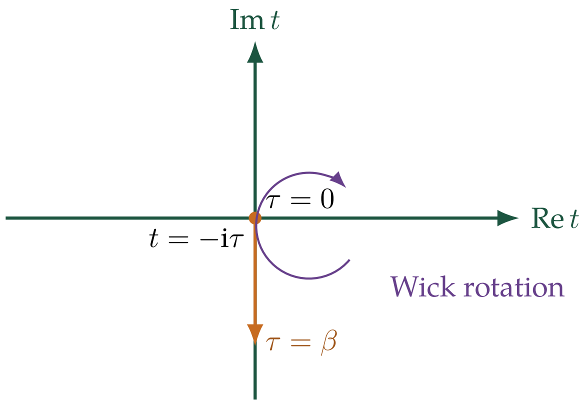


图 **15.2:** Real time lies on the horizontal axis. Thermal correlations evolve down the imaginary-time segment $t = -\mathrm{i}\tau$, $0 \le \tau \le \beta$.

**What cyclicity can and cannot do**

Cyclicity is sufficient to reduce $Z'$ to a single expectation value. For the second derivative, two copies of $A$ occur at different imaginary times, and cyclicity cannot in general turn the integral into $\beta\langle A^2\rangle$. That simplification is valid only when the relevant operators commute.

## 15.5 The Lie algebra of a closed matrix group

**Definition 15.4: Linear Lie group and its Lie algebra**

A *linear Lie group* over $\mathbb{K}$ is a subgroup

$$G \leq \mathrm{GL}(n, \mathbb{K})$$

that is closed in the matrix topology. Its Lie algebra is

$$\boxed{\mathrm{Lie}(G) := \left\{X \in \mathrm{Mat}_n(\mathbb{K}) : \mathrm{e}^{tX} \in G \text{ for every } t \in \mathbb{R}\right\}.} \tag{15.23}$$

We often abbreviate $\mathfrak{g} = \mathrm{Lie}(G)$.

The definition says exactly that $X \in \mathfrak{g}$ when the one-parameter subgroup generated by $X$ lies entirely inside $G$. Notice that the parameter $t$ is real even when $\mathbb{K} = \mathbb{C}$.

**Theorem 15.6: The set $\mathrm{Lie}(G)$ is a Lie algebra**

Let $G \leq \mathrm{GL}(n, \mathbb{K})$ be closed. Then $\mathfrak{g} = \mathrm{Lie}(G)$ is a real vector subspace of $\mathrm{Mat}_n(\mathbb{K})$, is closed in the matrix topology, and is closed under the commutator bracket

$$[X, Y] = XY - YX.$$

Hence $\mathfrak{g}$ is a real matrix Lie algebra.

证明. **Addition.** Suppose $X, Y \in \mathfrak{g}$. For every $t \in \mathbb{R}$ and $m \in \mathbb{N}$,

$$\mathrm{e}^{tX/m} \in G, \qquad \mathrm{e}^{tY/m} \in G,$$

so, because $G$ is a subgroup,

$$\left(\mathrm{e}^{tX/m}\mathrm{e}^{tY/m}\right)^m \in G.$$

The Lie–Trotter formula gives

$$\mathrm{e}^{t(X+Y)} = \lim_{m\to\infty} \left(\mathrm{e}^{tX/m}\mathrm{e}^{tY/m}\right)^m.$$

Closedness of $G$ implies $\mathrm{e}^{t(X+Y)} \in G$. Since this is true for every $t$, $X + Y \in \mathfrak{g}$.

**Real scalar multiplication.** If $\lambda \in \mathbb{R}$, then

$$\mathrm{e}^{t(\lambda X)} = \mathrm{e}^{(t\lambda)X} \in G \qquad (t \in \mathbb{R}),$$

so $\lambda X \in \mathfrak{g}$.

**Closedness of $\mathfrak{g}$.** Let $X_j \in \mathfrak{g}$ and $X_j \to X$. For every fixed $t$, continuity of the exponential gives

$$\mathrm{e}^{tX_j} \longrightarrow \mathrm{e}^{tX}.$$

Every $\mathrm{e}^{tX_j}$ lies in $G$, and $G$ is closed; hence $\mathrm{e}^{tX} \in G$. Therefore $X \in \mathfrak{g}$.

**Closure under brackets.** For $X, Y \in \mathfrak{g}$, define

$$C(t) := \mathrm{e}^{tY} X \mathrm{e}^{-tY}.$$

We first check that $C(t) \in \mathfrak{g}$. For every $s \in \mathbb{R}$, equivariance of the exponential gives

$$\mathrm{e}^{sC(t)} = \mathrm{e}^{tY} \mathrm{e}^{sX} \mathrm{e}^{-tY} \in G.$$

Thus $C(t) \in \mathfrak{g}$. Since $\mathfrak{g}$ is a vector space,

$$\frac{C(t) - C(0)}{t} \in \mathfrak{g} \qquad (t \neq 0).$$

Since $\mathfrak{g}$ is closed, the limit belongs to $\mathfrak{g}$:

$$\begin{aligned} C'(0) &= \frac{\mathrm{d}}{\mathrm{d}t} \left( \mathrm{e}^{tY} X \mathrm{e}^{-tY} \right) \Big|_{t=0} \\ &= YX - XY \\ &= -[X, Y]. \end{aligned}$$

Finally, real scalar closure gives $[X, Y] \in \mathfrak{g}$. ■

**Lemma 15.2: Invariance under group conjugation**

For $A \in G$ and $X \in \mathfrak{g}$,

$$\boxed{AXA^{-1} \in \mathfrak{g}.} \tag{15.24}$$

证明. For every $t \in \mathbb{R}$,

$$\mathrm{e}^{tAXA^{-1}} = A\mathrm{e}^{tX} A^{-1}.$$

All three factors on the right lie in $G$, so the product lies in $G$. Definition (15.23) now gives $AXA^{-1} \in \mathfrak{g}$. ■

**Analogy with a normal subgroup**

Equation (15.24) resembles the defining condition for a normal subgroup: $\mathfrak{g}$ is invariant under conjugation by elements of $G$. Nevertheless $\mathfrak{g}$ is not a subgroup of $G$. Its operation as a vector space is addition, its zero is the zero matrix, and usually its elements are not group elements at all. By contrast, the identity of $G$ is $I$, and the group operation is matrix multiplication.

**Definition 15.5: Complex linear Lie group**

Suppose $G \leq \mathrm{GL}(n, \mathbb{C})$. We call $G$ a *complex linear Lie group* if $\mathrm{Lie}(G)$ is a complex vector subspace of $\mathrm{Mat}_n(\mathbb{C})$. Equivalently,

$$X \in \mathrm{Lie}(G) \quad \Longrightarrow \quad \mathrm{i}X \in \mathrm{Lie}(G).$$

Without this extra property, $\mathrm{Lie}(G)$ is regarded as a real Lie algebra even though its matrices have complex entries.

The groups $\mathrm{GL}(n, \mathbb{C})$ and $\mathrm{SL}(n, \mathbb{C})$ are complex linear Lie groups. The unitary groups are not: multiplying a skew-Hermitian matrix by i makes it Hermitian rather than skew-Hermitian.

### 15.5.1 Exponentials lie in the identity component

Let $G^\circ$ denote the connected component of $I$ in $G$. If $X \in \mathfrak{g}$, the path

$$[0,1] \longrightarrow G, \qquad s \longmapsto \mathrm{e}^{sX},$$

joins $I$ to $\mathrm{e}^X$. Therefore

$$\boxed{\mathrm{e}^X \in G^\circ \quad \text{for every } X \in \mathfrak{g}.} \tag{15.25}$$

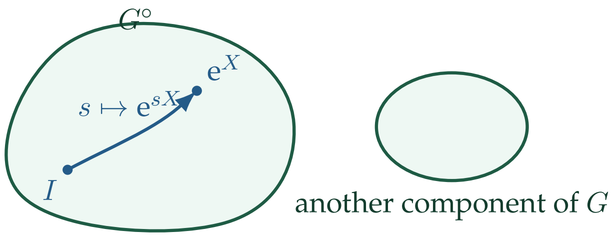


图 **15.3:** Every exponential generated by $\mathrm{Lie}(G)$ lies in the identity component.

**Proposition 15.4: Abelian groups and Abelian Lie algebras**

If $G$ is Abelian, then $\mathrm{Lie}(G)$ is Abelian.

证明. For $X, Y \in \mathfrak{g}$, the group elements $\mathrm{e}^{tX}$ and $\mathrm{e}^{sY}$ commute for every $s, t$. Hence the group commutator

$$K(t,s) := \mathrm{e}^{tX}\mathrm{e}^{sY}\mathrm{e}^{-tX}\mathrm{e}^{-sY}$$

is identically $I$. Differentiating once in each variable at $(t, s) = (0, 0)$ gives

$$0 = \left.\frac{\partial^2 K}{\partial t\,\partial s}\right|_{(0,0)} = XY - YX = [X, Y].$$

Thus all brackets in $\mathfrak{g}$ vanish. ■

The converse needs connectedness. It will follow in Section 15.9: if $G$ is connected and $\mathrm{Lie}(G)$ is Abelian, then $G$ is Abelian.

## 15.6 Classical groups and their Lie algebras

The definition (15.23), together with the identities already proved, quickly recovers the familiar classical matrix Lie algebras.

### 15.6.1 General and special linear groups

**Example 15.2:** $\mathrm{GL}(n, \mathbb{C})$ **and** $\mathrm{GL}(n, \mathbb{R})$

Every matrix exponential is invertible. Therefore

$$\mathrm{Lie}(\mathrm{GL}(n, \mathbb{C})) = \mathfrak{gl}(n, \mathbb{C}) = \mathrm{Mat}_n(\mathbb{C}),$$

which is a complex vector space, whereas

$$\mathrm{Lie}(\mathrm{GL}(n, \mathbb{R})) = \mathfrak{gl}(n, \mathbb{R}) = \mathrm{Mat}_n(\mathbb{R}),$$

which is a real vector space. If $X$ is real, $\mathrm{e}^{tX}$ is real for real $t$; there is no reason for $\mathrm{e}^{\mathrm{i}tX}$ to be real.

**Example 15.3:** $\mathrm{SL}(n, \mathbb{K})$

By (15.6),

$$\det(\mathrm{e}^{tX}) = \mathrm{e}^{t \operatorname{tr} X}.$$

If $\operatorname{tr} X = 0$, then $\det(\mathrm{e}^{tX}) = 1$ for every $t$, so $X \in \mathrm{Lie}(\mathrm{SL}(n, \mathbb{K}))$.
Conversely, if $\mathrm{e}^{tX} \in \mathrm{SL}(n, \mathbb{K})$ for every real $t$, then

$$1 = \mathrm{e}^{t \operatorname{tr} X}.$$

Differentiating at $t = 0$ gives $\operatorname{tr} X = 0$. Hence

$$\boxed{\mathrm{Lie}(\mathrm{SL}(n, \mathbb{K})) = \mathfrak{sl}(n, \mathbb{K}) = \{X \in \mathrm{Mat}_n(\mathbb{K}) : \operatorname{tr} X = 0\}.}$$

For $\mathbb{K} = \mathbb{C}$, this is a complex linear Lie algebra; for $\mathbb{K} = \mathbb{R}$, it is real.

### 15.6.2 Unitary and orthogonal groups

**Example 15.4: The unitary Lie algebra**

For $X \in \mathrm{Mat}_n(\mathbb{C})$,

$$\begin{aligned} \mathrm{e}^{tX} \in U(n) \quad \text{for every } t &\iff (\mathrm{e}^{tX})^\dagger \mathrm{e}^{tX} = I \quad \text{for every } t \\ &\iff \mathrm{e}^{tX^\dagger} \mathrm{e}^{tX} = I \quad \text{for every } t. \end{aligned}$$

If $X^\dagger = -X$, the two exponents commute and the last product equals $\mathrm{e}^{-tX}\mathrm{e}^{tX} = I$. Conversely, differentiating the unitary condition at $t = 0$ gives

$$X^\dagger + X = 0.$$

Therefore

$$\boxed{\mathfrak{u}(n) = \mathrm{Lie}(U(n)) = \{X \in \mathrm{Mat}_n(\mathbb{C}) : X^\dagger = -X\}.} \tag{15.26}$$

Adding the determinant-one condition adds the infinitesimal trace-zero condition:

$$\boxed{\mathfrak{su}(n) = \mathrm{Lie}(SU(n)) = \{X : X^\dagger = -X,\ \mathrm{tr}\, X = 0\}.} \tag{15.27}$$

Neither $U(n)$ nor $SU(n)$ is a complex linear Lie group. Indeed, if $X^\dagger = -X$, then

$$(\mathrm{i}X)^\dagger = -\mathrm{i}X^\dagger = \mathrm{i}X,$$

so $\mathrm{i}X$ is Hermitian, not skew-Hermitian, unless $X = 0$.

**Example 15.5: The orthogonal Lie algebra**

Exactly the same calculation with transpose in place of adjoint gives

$$\boxed{\mathfrak{o}(n) = \mathrm{Lie}(O(n)) = \{X \in \mathrm{Mat}_n(\mathbb{R}) : X^T = -X\}.} \tag{15.28}$$

A real skew-symmetric matrix has zero diagonal, hence zero trace. Therefore the additional condition $\det A = 1$ does not reduce the Lie algebra:

$$\boxed{\mathrm{Lie}(SO(n)) = \mathfrak{so}(n) = \mathfrak{o}(n).} \tag{15.29}$$

Thus $O(n)$ and its identity component $SO(n)$ have the same infinitesimal directions, even though $O(n)$ has two connected components.

| **Group** $G$ | $\mathrm{Lie}(G)$ | **scalar field of the Lie algebra** |
|---|---|---|
| $\mathrm{GL}(n,\mathbb{C})$ | all $X \in \mathrm{Mat}_n(\mathbb{C})$ | complex |
| $\mathrm{SL}(n,\mathbb{C})$ | $\mathrm{tr}\, X = 0$ | complex |
| $\mathrm{GL}(n,\mathbb{R})$ | all $X \in \mathrm{Mat}_n(\mathbb{R})$ | real |
| $\mathrm{SL}(n,\mathbb{R})$ | $\mathrm{tr}\, X = 0$ | real |
| $U(n)$ | $X^\dagger = -X$ | real |
| $SU(n)$ | $X^\dagger = -X,\ \mathrm{tr}\, X = 0$ | real |
| $O(n)$, $SO(n)$ | $X^T = -X,\ X$ real | real |

表 **15.1:** Classical matrix groups and their Lie algebras.

### 15.6.3 The physics convention

Mathematicians usually put skew-Hermitian generators in $\mathfrak{u}(n)$, so a one-parameter unitary group is written $\mathrm{e}^{tX}$. In physics, one normally prefers Hermitian generators and writes

$$U(t) = \mathrm{e}^{\mathrm{i}tH}.$$

The two conventions differ by the factor i:

$$X = \mathrm{i}H, \qquad X^\dagger = -X \iff H^\dagger = H.$$

Accordingly, in the physics convention one may identify

$$\mathfrak{u}(n)_{\text{phys}} = \{H : H^\dagger = H\},$$

and

$$\mathfrak{su}(n)_{\text{phys}} = \{H : H^\dagger = H,\ \mathrm{tr}\, H = 0\}.$$

One must then also transport the bracket. If $X = \mathrm{i}H$ and $Y = \mathrm{i}K$, the skew-Hermitian commutator corresponds to the Hermitian bracket

$$[H, K]_{\text{phys}} := \mathrm{i}(HK - KH)$$

up to the sign convention chosen by the author. Keeping track of this factor prevents apparent disagreements between mathematics and physics texts.

## 15.7 The exponential map of a linear Lie group

**Definition 15.6: Group exponential map**

For a linear Lie group $G$ with $\mathfrak{g} = \mathrm{Lie}(G)$, the *exponential map* of $G$ is the restriction

$$\exp : \mathfrak{g} \longrightarrow G, \qquad X \longmapsto \mathrm{e}^X. \tag{15.30}$$

The map need be neither injective nor surjective globally, although it is always locally bijective at 0.

**Example 15.6: Failure of injectivity for $SU(2)$**

Let
$$X = \begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix} \in \mathfrak{su}(2).$$
Then
$$\mathrm{e}^{0X} = I, \qquad \mathrm{e}^{2\pi X} = \begin{pmatrix} \mathrm{e}^{2\pi \mathrm{i}} & 0 \\ 0 & \mathrm{e}^{-2\pi \mathrm{i}} \end{pmatrix} = I.$$
Since $0 \neq 2\pi X$, the map $\exp : \mathfrak{su}(2) \to SU(2)$ is not injective.

**Example 15.7: Failure of surjectivity for $\mathrm{SL}(2,\mathbb{C})$**

Consider
$$A = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$
Its determinant is $1$, so $A \in \mathrm{SL}(2,\mathbb{C})$. Its only eigenvalue is $-1$, with algebraic multiplicity two, but
$$A + I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$
has a one-dimensional kernel. Hence $A$ is not diagonalizable.
Suppose, for contradiction, that $A = \mathrm{e}^X$ for some $X \in \mathfrak{sl}(2,\mathbb{C})$. If $\mu_1, \mu_2$ are the eigenvalues of $X$, spectral mapping gives $\mathrm{e}^{\mu_1} = \mathrm{e}^{\mu_2} = -1$. Thus
$$\mu_j = (2k_j + 1)\pi \mathrm{i} \qquad (k_j \in \mathbb{Z}).$$
Because $\operatorname{tr} X = 0$, $\mu_2 = -\mu_1$; in particular the two eigenvalues are distinct, since neither can be zero. Therefore $X$ is diagonalizable. In a basis diagonalizing $X$,
$$\mathrm{e}^X = \operatorname{diag}(-1,-1) = -I,$$
contradicting $A \neq -I$. Consequently
$$\exp : \mathfrak{sl}(2,\mathbb{C}) \longrightarrow \mathrm{SL}(2,\mathbb{C})$$
is not surjective.

### 15.7.1 Local bijectivity on a closed matrix group

**Theorem 15.7: Local exponential theorem**

Let $G \leq \mathrm{GL}(n,\mathbb{K})$ be a closed matrix group and $\mathfrak{g} = \mathrm{Lie}(G)$. There are neighbourhoods $\mathcal{U}_G$ of $0$ in $\mathfrak{g}$ and $\mathcal{V}_G$ of $I$ in $G$ such that
$$\exp : \mathcal{U}_G \longrightarrow \mathcal{V}_G \tag{15.31}$$
is a smooth bijection with smooth inverse log. Equivalently, every element of $G$ sufficiently close to $I$ has a unique sufficiently small logarithm belonging to $\mathfrak{g}$.

证明. The ambient exponential is a smooth bijection from a neighbourhood $\mathcal{U} \subset \mathrm{Mat}_n(\mathbb{K})$ of $0$ onto a neighbourhood $\mathcal{V} \subset \mathrm{GL}(n, \mathbb{K})$ of $I$, with inverse given by the convergent logarithm series. By the closed-subgroup theorem proved in Chapter 13, a closed matrix group is an embedded submanifold, and its tangent space at $I$ is precisely the space of infinitesimal one-parameter subgroups. In the present notation that space is $\mathfrak{g}$. The differential of the exponential at $0$ is

$$(\mathrm{d}\exp)_0(X) = \frac{\mathrm{d}}{\mathrm{d}t} \mathrm{e}^{tX}\Big|_{t=0} = X,$$

so its restriction from $\mathfrak{g}$ to $T_I G$ is the identity. The inverse-function theorem on these equal-dimensional manifolds therefore gives neighbourhoods $\mathcal{U}_G$ and $\mathcal{V}_G$ for which (15.31) is a diffeomorphism.

The direct tangent-space verification $T_I G = \mathfrak{g}$, which also removes any possible ambiguity in this use of the closed-subgroup theorem, is given below. ■

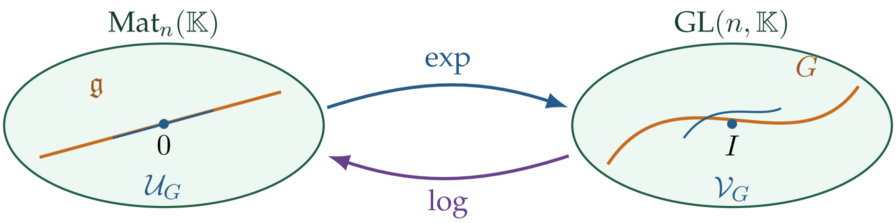


图 **15.4:** Restriction of the ambient local exponential chart to $\mathfrak{g}$ and $G$.

## 15.8 The tangent-space interpretation

**Definition 15.7: Tangent space to a matrix group**

Let $G \leq \mathrm{GL}(n, \mathbb{K})$ be a linear Lie group and let $A \in G$. The tangent space at $A$ is

$$T_A G := \{\dot{c}(0) : c : (-\varepsilon, \varepsilon) \to G \text{ is smooth and } c(0) = A\}. \tag{15.32}$$

Thus a tangent vector is the velocity of a smooth curve in $G$.

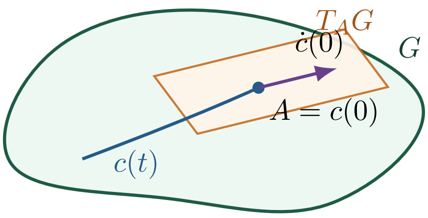


图 **15.5:** A smooth curve in $G$ and its tangent vector at $A$.

**Theorem 15.8: Lie algebra equals the tangent space at the identity**

For every linear Lie group $G$,

$$\boxed{T_I G = \mathrm{Lie}(G).} \tag{15.33}$$

证明. Lie$(G) \subseteq T_I G$. If $X \in$ Lie$(G)$, the curve

$$c_X(t) = \mathrm{e}^{tX}$$

lies in $G$, satisfies $c_X(0) = I$, and has $\dot{c}_X(0) = X$. Hence $X \in T_I G$.

$T_I G \subseteq$ Lie$(G)$. Let $X \in T_I G$. Then there is a smooth curve

$$c : (-\varepsilon, \varepsilon) \to G, \qquad c(0) = I, \qquad \dot{c}(0) = X.$$

For sufficiently small $t$, $c(t)$ lies in the neighbourhood $\mathcal{V}_G$ from the local exponential theorem above. Therefore $\log c(t) \in \mathfrak{g}$. Since $\mathfrak{g}$ is a vector space,

$$\frac{\log c(t)}{t} \in \mathfrak{g} \qquad (t \neq 0).$$

Now expand the logarithm:

$$\log c(t) = (c(t) - I) - \frac{1}{2}(c(t) - I)^2 + \cdots .$$

Because $c(t) - I = tX + o(t)$, every term of degree at least two is $o(t)$. Thus

$$\lim_{t \to 0} \frac{\log c(t)}{t} = \lim_{t \to 0} \frac{c(t) - I}{t} = \dot{c}(0) = X.$$

The Lie algebra $\mathfrak{g}$ is closed, so $X \in \mathfrak{g}$. ■

**Proposition 15.5: Tangent spaces at arbitrary points**

For every $A \in G$,

$$\boxed{T_A G = A\, T_I G = A\mathfrak{g}.} \tag{15.34}$$

Equivalently, right translation gives $T_A G = \mathfrak{g} A$.

证明. If $c(0) = I$ and $\dot{c}(0) = X$, then $\bar{c}(t) := Ac(t)$ satisfies

$$\bar{c}(0) = A, \qquad \dot{\bar{c}}(0) = AX.$$

This proves $A\mathfrak{g} \subseteq T_A G$. Conversely, if $\bar{c}(0) = A$, then $c(t) := A^{-1}\bar{c}(t)$ is a curve through $I$, and

$$\dot{c}(0) = A^{-1}\dot{\bar{c}}(0) \in \mathfrak{g}.$$

Thus $\dot{\bar{c}}(0) \in A\mathfrak{g}$, proving equality. ■

**Definition 15.8: Dimension of a linear Lie group**

The dimension of $G$ is

$$\dim G := \dim_{\mathbb{R}} \mathrm{Lie}(G). \tag{15.35}$$

Even when the matrices are complex, the Lie algebra is counted as a real vector space unless $G$ has explicitly been declared a complex Lie group and a complex dimension is requested.

Since every tangent space is a translate of $T_I G$, all tangent spaces have the same dimension. The closed-subgroup theorem therefore realizes $G$ as an embedded submanifold of $\mathrm{Mat}_n(\mathbb{K})$ of dimension $\dim_{\mathbb{R}} \mathfrak{g}$.

## 15.9 Connected groups are generated by exponentials

A Lie group is locally Euclidean and therefore locally path connected. Consequently, a connected Lie group is path connected. If $G$ is connected and $A \in G$, we may choose a continuous path

$$c : [0,1] \longrightarrow G, \qquad c(0) = I, \qquad c(1) = A.$$

Uniform continuity of $c$ allows the path to be divided into sufficiently short pieces whose relative increments lie in the local exponential neighbourhood.

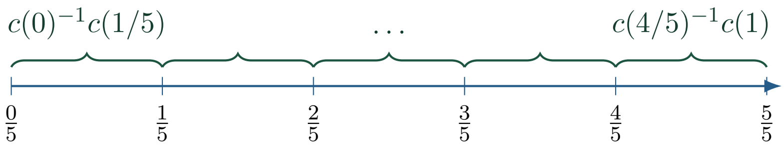


图 **15.6:** Divide a path into pieces whose relative changes are all close to the identity.

**Theorem 15.9: Finite product of exponentials**

If $G$ is a connected linear Lie group, then every $A \in G$ can be written

$$\boxed{A = \mathrm{e}^{X_1}\mathrm{e}^{X_2}\cdots\mathrm{e}^{X_m}, \qquad X_j \in \mathrm{Lie}(G).} \tag{15.36}$$

证明. Choose a path $c$ from $I$ to $A$. Let $\mathcal{V}_G$ be the neighbourhood of $I$ on which the group logarithm exists. Because the map

$$(s,t) \longmapsto c(s)^{-1}c(t)$$

is continuous on the compact square $[0,1]^2$, it is uniformly continuous. We may therefore choose $m$ large enough that

$$c((j-1)/m)^{-1}c(j/m) \in \mathcal{V}_G \qquad (j = 1, \ldots, m).$$

By the local exponential theorem, for each $j$ there is an $X_j \in \mathfrak{g}$ such that

$$c((j-1)/m)^{-1}c(j/m) = \mathrm{e}^{X_j}.$$

Multiplying these equations in order makes all intermediate path values cancel:

$$\begin{aligned} A &= c(0)^{-1}c(1) \\ &= \left[c(0)^{-1}c(1/m)\right]\left[c(1/m)^{-1}c(2/m)\right]\cdots\left[c((m-1)/m)^{-1}c(1)\right] \\ &= \mathrm{e}^{X_1}\mathrm{e}^{X_2}\cdots\mathrm{e}^{X_m}. \end{aligned}$$

■

**One exponential versus a product**

Connectedness guarantees a *finite product* of exponentials. It does not guarantee that $A = \mathrm{e}^X$ for one $X$. The defective matrix in $\mathrm{SL}(2, \mathbb{C})$ above shows why the distinction is essential.

**Theorem 15.10: Connected group with Abelian Lie algebra**

If $G$ is connected and $\mathrm{Lie}(G)$ is Abelian, then $G$ is Abelian.

证明. If $X, Y \in \mathfrak{g}$, then $[X, Y] = 0$, so

$$\mathrm{e}^X \mathrm{e}^Y = \mathrm{e}^{X+Y} = \mathrm{e}^Y \mathrm{e}^X.$$

Write arbitrary $A, B \in G$ as products of exponentials:

$$A = \mathrm{e}^{X_1} \cdots \mathrm{e}^{X_m}, \qquad B = \mathrm{e}^{Y_1} \cdots \mathrm{e}^{Y_r}.$$

Every $X_i$ commutes with every $Y_j$, so all the exponential factors commute. Reordering them gives $AB = BA$. ■

Combining the two implications just proved, we obtain

$$G \text{ Abelian} \Longrightarrow \mathrm{Lie}(G) \text{ Abelian}$$

without any connectedness assumption, and

$$\mathrm{Lie}(G) \text{ Abelian} \Longrightarrow G \text{ Abelian}$$

when $G$ is connected.

## 15.10 When one exponential is enough

### 15.10.1 Surjectivity for $SU(n)$

**Theorem 15.11: Every special unitary matrix is an exponential**

For every $A \in SU(n)$, there exists $X \in \mathfrak{su}(n)$ such that

$$A = \mathrm{e}^X.$$

Thus $\exp : \mathfrak{su}(n) \to SU(n)$ is surjective.

证明. A unitary matrix is normal, so the spectral theorem gives a unitary matrix $S$ and phases $\theta_1, \dots, \theta_n \in \mathbb{R}$ such that

$$A = SDS^\dagger, \qquad D = \mathrm{diag}(\mathrm{e}^{\mathrm{i}\theta_1}, \dots, \mathrm{e}^{\mathrm{i}\theta_n}).$$

Because $\det A = 1$,

$$1 = \det D = \mathrm{e}^{\mathrm{i}(\theta_1 + \cdots + \theta_n)}.$$

Therefore
$$\theta_1 + \cdots + \theta_n = 2\pi k \qquad \text{for some } k \in \mathbb{Z}.$$
Replace, for example, $\theta_n$ by
$$\theta_n' := \theta_n - 2\pi k.$$
This does not change $e^{i\theta_n}$, but now
$$\theta_1 + \cdots + \theta_{n-1} + \theta_n' = 0.$$
Set
$$\Theta := \operatorname{diag}(\theta_1, \ldots, \theta_{n-1}, \theta_n'), \qquad X := iS\Theta S^\dagger.$$
Then
$$X^\dagger = -iS\Theta S^\dagger = -X, \qquad \operatorname{tr} X = i \operatorname{tr} \Theta = 0,$$
so $X \in \mathfrak{su}(n)$. Finally,
$$e^X = Se^{i\Theta}S^\dagger = SDS^\dagger = A.$$
■

**The phase correction that must not be skipped**

Choosing all phases in $[0, 2\pi)$ only implies that their sum is an integer multiple of $2\pi$, not literally zero. Subtracting that multiple from one phase leaves the unitary eigenvalue unchanged and makes the logarithm traceless. This small adjustment is essential for the generator to lie in $\mathfrak{su}(n)$.

### 15.10.2 Surjectivity for $SO(n)$

**Theorem 15.12: Every special orthogonal matrix is an exponential**

For every $A \in SO(n)$, there exists $X \in \mathfrak{so}(n)$ such that
$$A = e^X.$$

证明. The real spectral theorem for orthogonal matrices gives an orthogonal matrix $Q$ for which $Q^T AQ$ is block diagonal with $2 \times 2$ rotation blocks
$$R(\theta_j) = \begin{pmatrix} \cos\theta_j & -\sin\theta_j \\ \sin\theta_j & \cos\theta_j \end{pmatrix}$$
and $1 \times 1$ blocks $1$ or $-1$. Since $\det A = 1$, the number of $-1$ blocks is even. Pair two $-1$ blocks and regard their direct sum as $R(\pi)$. We may therefore write
$$Q^T AQ = \operatorname{diag}\big(R(\theta_1), \ldots, R(\theta_r), I_{n-2r}\big).$$
Let
$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Since $J^2 = -I_2$, separating the exponential series into even and odd powers gives

$$e^{\theta J} = I_2 \cos\theta + J \sin\theta = R(\theta).$$

Now define

$$X := Q \operatorname{diag}\big(\theta_1 J, \ldots, \theta_r J, 0_{n-2r}\big) Q^T.$$

Every block $\theta_j J$ is real and skew-symmetric; hence $X^T = -X$, so $X \in \mathfrak{so}(n)$. Moreover,

$$e^X = Q \operatorname{diag}\big(R(\theta_1), \ldots, R(\theta_r), I\big) Q^T = A.$$

■

**Why a local argument is not enough here**

It is tempting to embed $SO(n)$ in $SU(n)$, write $A = e^X$, and infer from $AA^T = I$ that $X + X^T = 0$. That inference is valid only in a neighbourhood where the logarithm is unique, or when the required commutativity has been established. The rotation-block construction above is global and directly produces a real skew-symmetric logarithm.

**Theorem 15.13: Connected compact groups have surjective exponential**

If $G$ is a connected compact linear Lie group, then

$$G = \exp(\mathrm{Lie}(G)). \tag{15.37}$$

证明. The maximal-torus theorem says that every element $A$ of a compact connected Lie group lies in some maximal torus $T \leq G$. A torus is isomorphic to

$$U(1)^r = \mathbb{R}^r/(2\pi\mathbb{Z})^r$$

for some $r$, and its exponential map

$$\mathrm{Lie}(T) \cong \mathbb{R}^r \longrightarrow T$$

is surjective, coordinate by coordinate:

$$(\theta_1, \ldots, \theta_r) \longmapsto (e^{i\theta_1}, \ldots, e^{i\theta_r}).$$

Thus $A = e^X$ for some $X \in \mathrm{Lie}(T) \subseteq \mathrm{Lie}(G)$. ■

**Role of compactness**

The theorem explains the surjectivity for $SU(n)$ and $SO(n)$, both of which are compact and connected. Compactness cannot be omitted: $\mathrm{SL}(2, \mathbb{C})$ is connected but the earlier defective matrix is not one exponential.

## 15.11 The identity component has the same Lie algebra

**Theorem 15.14: The identity component of a linear Lie group**

Let $G \leq \mathrm{GL}(n, \mathbb{K})$ be a linear Lie group. Then

1. $G^\circ$ is a closed normal subgroup of $G$, hence itself a linear Lie group;
2. $$\boxed{\mathrm{Lie}(G^\circ) = \mathrm{Lie}(G).} \tag{15.38}$$

证明. The fact that the identity component is a normal subgroup was proved topologically in Chapter 13: multiplication, inversion, and conjugation send connected sets to connected sets.

It remains useful to see directly why $G^\circ$ is closed in the matrix topology. Let $A_m \in G^\circ$ and suppose $A_m \to A$. Since $G$ is closed, $A \in G$. Furthermore,

$$A_m^{-1} A \longrightarrow I.$$

For sufficiently large $m$, the local exponential theorem gives

$$A_m^{-1} A = \mathrm{e}^X \qquad \text{for some } X \in \mathfrak{g}.$$

There is a path in $G$ from $I$ to $A_m$, because $A_m \in G^\circ$. The curve

$$s \longmapsto A_m \mathrm{e}^{sX}, \qquad 0 \leq s \leq 1,$$

then joins $A_m$ to $A_m \mathrm{e}^X = A$. Concatenating the two paths joins $I$ to $A$, so $A \in G^\circ$. Thus $G^\circ$ is closed.

Because $G^\circ \subseteq G$,
$$\mathrm{Lie}(G^\circ) \subseteq \mathrm{Lie}(G).$$

Conversely, take $X \in \mathrm{Lie}(G)$. For each fixed $t \in \mathbb{R}$, the path

$$[0, 1] \longrightarrow G, \qquad s \longmapsto \mathrm{e}^{stX},$$

joins $I$ to $\mathrm{e}^{tX}$. Hence $\mathrm{e}^{tX} \in G^\circ$ for every $t$, which says $X \in \mathrm{Lie}(G^\circ)$. This proves the reverse inclusion and (15.38). ■

**Example 15.8: $O(n)$ and $SO(n)$**

The determinant is continuous and takes only the values $\pm 1$ on $O(n)$. Therefore no path in $O(n)$ can join a determinant-$+1$ matrix to a determinant-$-1$ matrix. The identity component is

$$O(n)^\circ = SO(n),$$

and the theorem gives
$$\mathrm{Lie}(O(n)) = \mathrm{Lie}(SO(n)) = \mathfrak{so}(n).$$

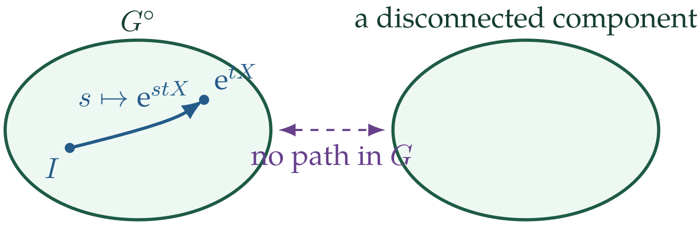


图 **15.7:** Every one-parameter subgroup through $I$ stays in $G^\circ$; the other components introduce no new infinitesimal directions.

**What the Lie algebra cannot see**

The Lie algebra records the geometry of $G$ arbitrarily close to the identity. It therefore determines the infinitesimal structure of $G^\circ$, but it does not record how many disconnected components the whole group has. More precisely, the discrete component group $G/G^\circ$ is invisible to Lie$(G)$. This is why $O(n)$ and $SO(n)$ have the same Lie algebra even though the first group has an additional determinant-$-1$ component.

**What to retain from this chapter**

1. The matrix exponential is the absolutely convergent series

$$e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}, \qquad \|e^X\|_F \le e^{\|X\|_F}.$$

It is always invertible, with inverse $e^{-X}$.

2. If $XY = YX$, then $e^{X+Y} = e^X e^Y$. For arbitrary matrices,

$$e^{X+Y} = \lim_{m\to\infty} \left(e^{X/m} e^{Y/m}\right)^m.$$

3. Similarity and determinant behave naturally:

$$e^{AXA^{-1}} = Ae^X A^{-1}, \qquad \det(e^X) = e^{\operatorname{tr} X}.$$

4. Near $I$,

$$\log A = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (A - I)^k$$

is the inverse of the exponential.

5. Every continuous one-parameter subgroup of $\mathrm{GL}(n, \mathbb{K})$ is uniquely $c(t) = e^{tX}$, with $X = \dot{c}(0)$.

6. For a closed matrix group,

$$\mathrm{Lie}(G) = \{X : e^{tX} \in G \text{ for all } t \in \mathbb{R}\}$$

is a closed real vector space, is closed under commutators, and is invariant under $X \mapsto AXA^{-1}$ for $A \in G$.

7. The same space is the tangent space at the identity:

$$T_I G = \mathrm{Lie}(G), \qquad T_A G = A\,\mathrm{Lie}(G).$$

8. The exponential is a local diffeomorphism from $\mathrm{Lie}(G)$ to $G$ near $0$ and $I$, but it need not be globally injective or surjective.

9. Every element of a connected linear Lie group is a finite product of exponentials. Every element is a *single* exponential when the group is connected and compact. In particular, the exponential maps onto $SU(n)$ and $SO(n)$.

10. The identity component is a closed normal linear Lie subgroup and

$$\mathrm{Lie}(G^\circ) = \mathrm{Lie}(G).$$